

Scalable Predictive Analytics of Student Burnout

A scalable GPU-accelerated, explainable machine learning for for early detection of student burnout risk.

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THE PROBLEM

Project Overview & Goal

Dataset

- Source: Kaggle – Student Mental Health & Burnout Dataset
- Scale: 1,000,000 student records
- Features: 20 variables across multiple domains

Feature Categories

- Academic: study hours, exam pressure, performance
- Behavioral: sleep hours, screen time, physical activity
- Psychological: stress level, anxiety score, depression score
- Socio-economic: financial stress, family expectation, social support

Data Characteristics

- High-dimensional and large-scale dataset
- Contains complex, non-linear relationships between variables
- Suitable for both exploratory analysis and predictive modeling

Project Objective

- Develop a scalable, GPU-accelerated, explainable AI system
- Predict burnout risk (Low / Medium / High)
- Identify key drivers of student burnout

Data Engineering & Feature Design

GPU-Accelerated Processing

- Implemented using NVIDIA RAPIDS (cuDF)
- CPU: 4.94s → GPU: 0.23s
- Achieved 21.67× speedup

Data Preparation

- Standardized features and handled missing values
- Applied outlier mitigation for stability
- Processed full 1M dataset without sampling

Feature Engineering

- Created 8 interaction-based features, key features:
 - Academic resilience
 - Study-sleep gap
 - Combined distress
 - Anxiety-pressure impact
 - Effort-Reward Disparity
 - Total Cognitive Load
 - Social Support Ratio
 - Socio-Economic Multiplier

Exploratory Data Analysis: Key Insights & Modeling Implications

- We analyzed relationships between key variables such as stress level, sleep duration, study workload, and financial stress to understand burnout behavior.
- Burnout remains relatively stable until a critical stress threshold (~8/10), after which it increases sharply.
- Students with low sleep and high study workload consistently exhibit the highest burnout risk.
- Financial stress remains persistently high across medium and high-risk groups, indicating a strong underlying factor.
- These patterns show that burnout is driven by interactions between variables rather than individual factors.
- This directly motivated the use of a non-linear model (XGBoost) to capture these complex relationships.

Machine Learning Approach for Burnout Prediction

Based on the patterns identified from data analysis, we formulated the problem as a multi-class classification task.

Objective

- Predict student burnout risk levels:
 - Low
 - Medium
 - High

Model Selection:

- XGBoost (Gradient Boosted Decision Trees)
- Chosen for its ability to capture non-linear relationships and feature interactions

Model Design:

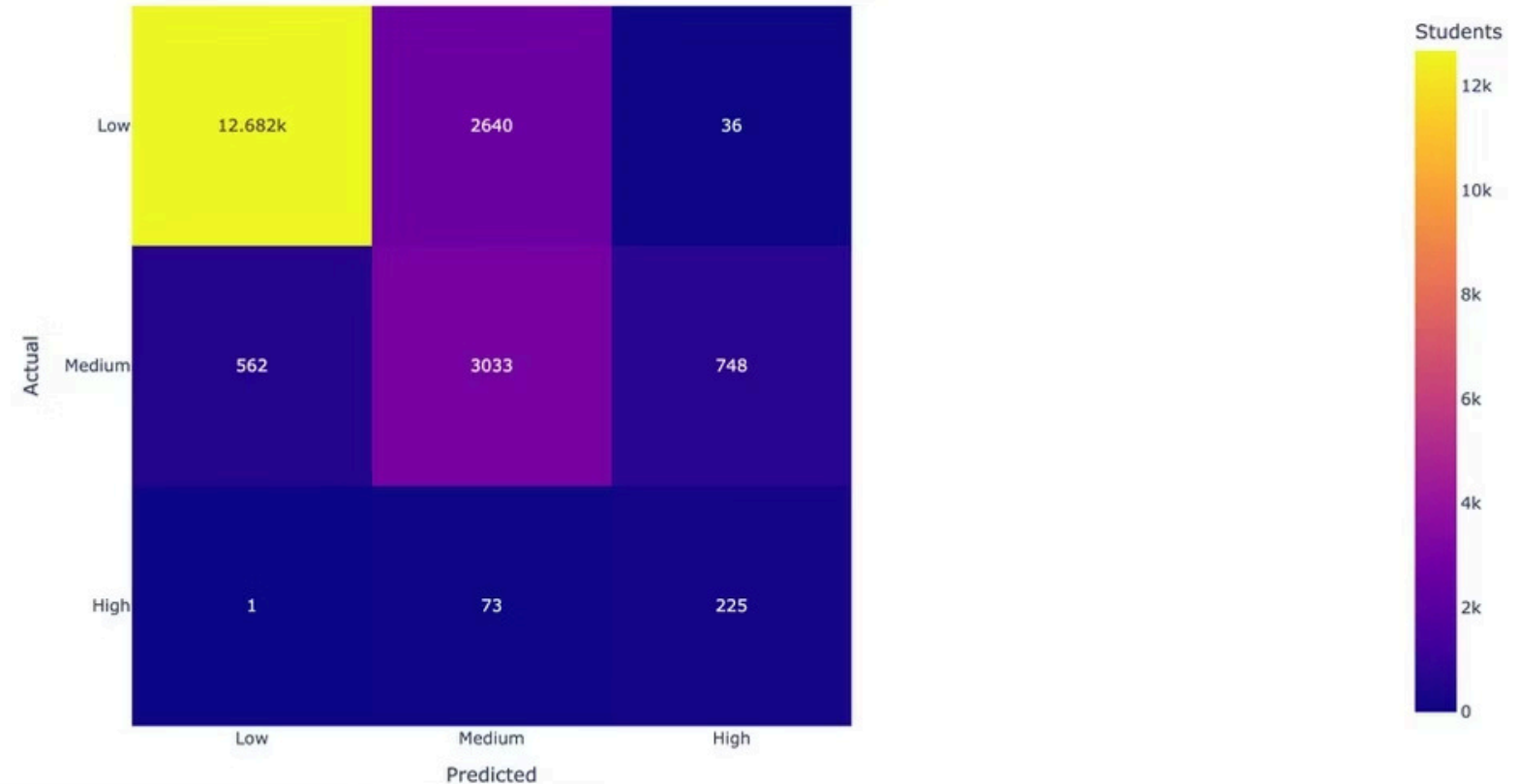
- Train-test split: 80% / 20%
- Used both original and engineered features
- Prevented data leakage by excluding target-related variables

Model Performance: Classification Results

The model demonstrates strong classification performance across all burnout risk levels.

- Most predictions are correctly classified (diagonal dominance)
- Minimal misclassification between adjacent risk categories
- High accuracy across Low, Medium, and High-risk groups

ML Visual 1: Confusion Matrix

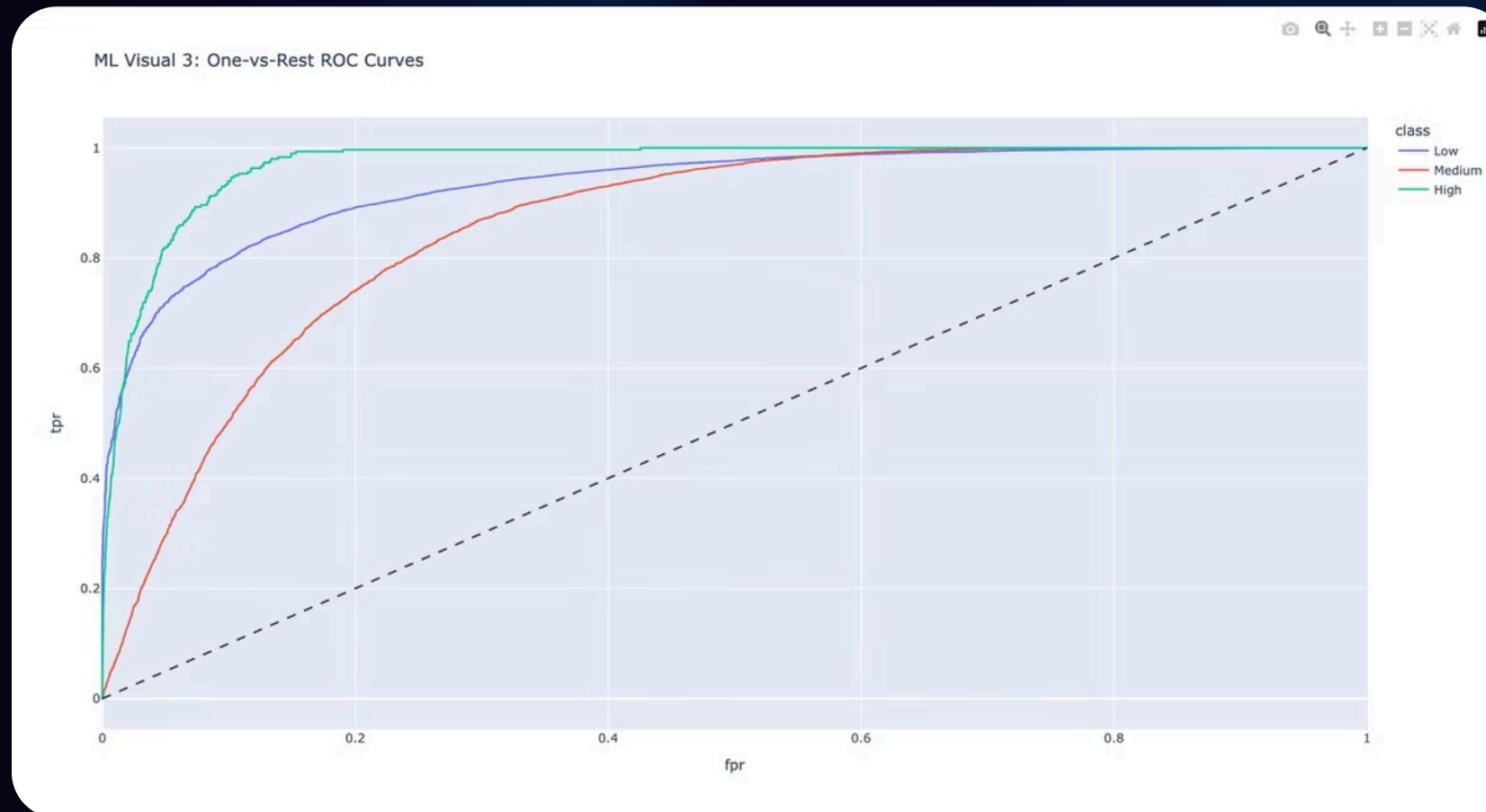


This confusion matrix shows that the model performs very well across all classes. Most predictions fall along the diagonal, indicating correct classification, with only minor misclassification between adjacent risk levels.

Model Evaluation: ROC and Precision-Recall Analysis

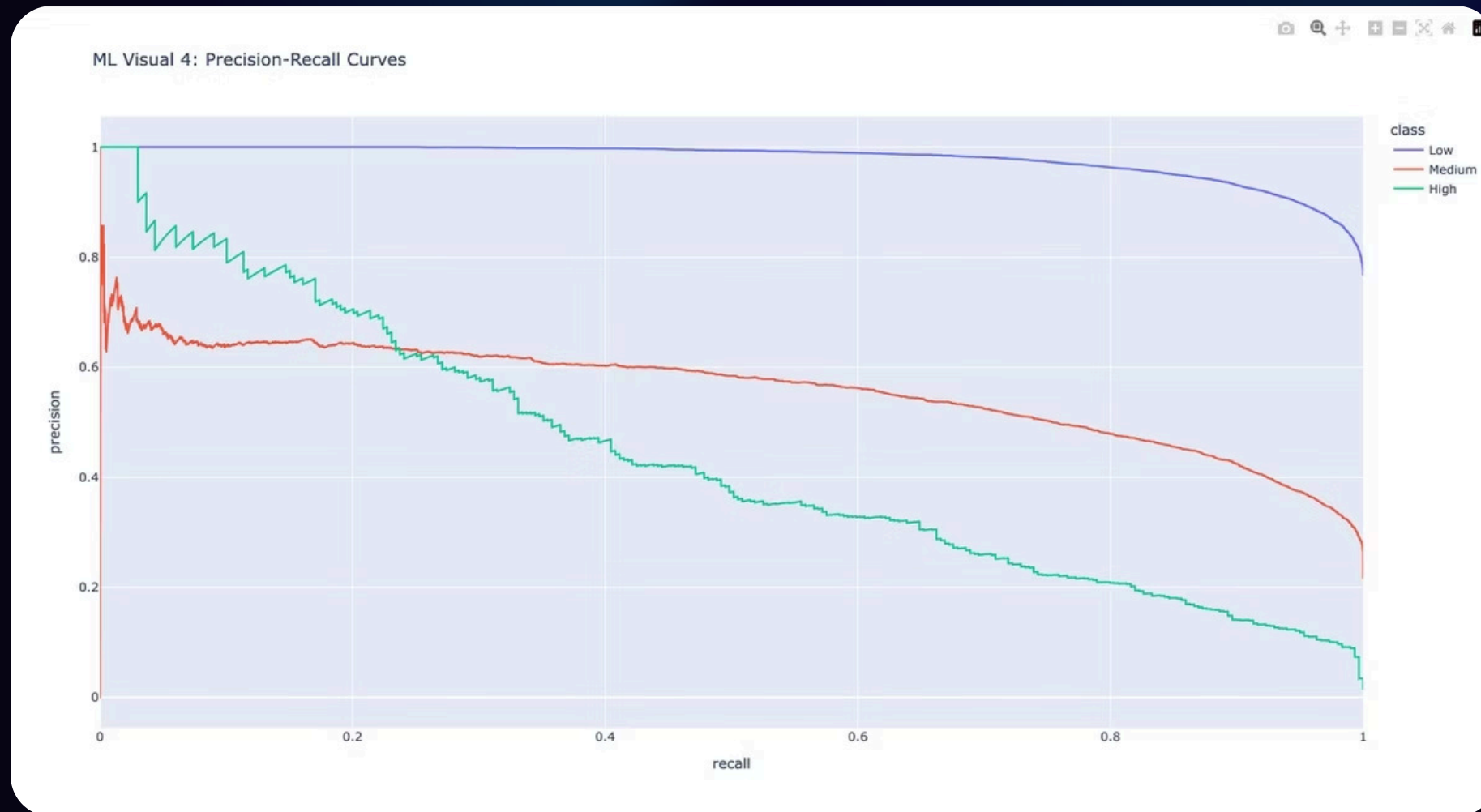
To further evaluate model performance, we analyzed ROC and Precision-Recall curves.

- ROC curves show strong separation between burnout risk classes
- High true positive rate with low false positive rate
- Consistent performance across Low, Medium, and High risk groups



Model Evaluation: Precision-Recall Analysis

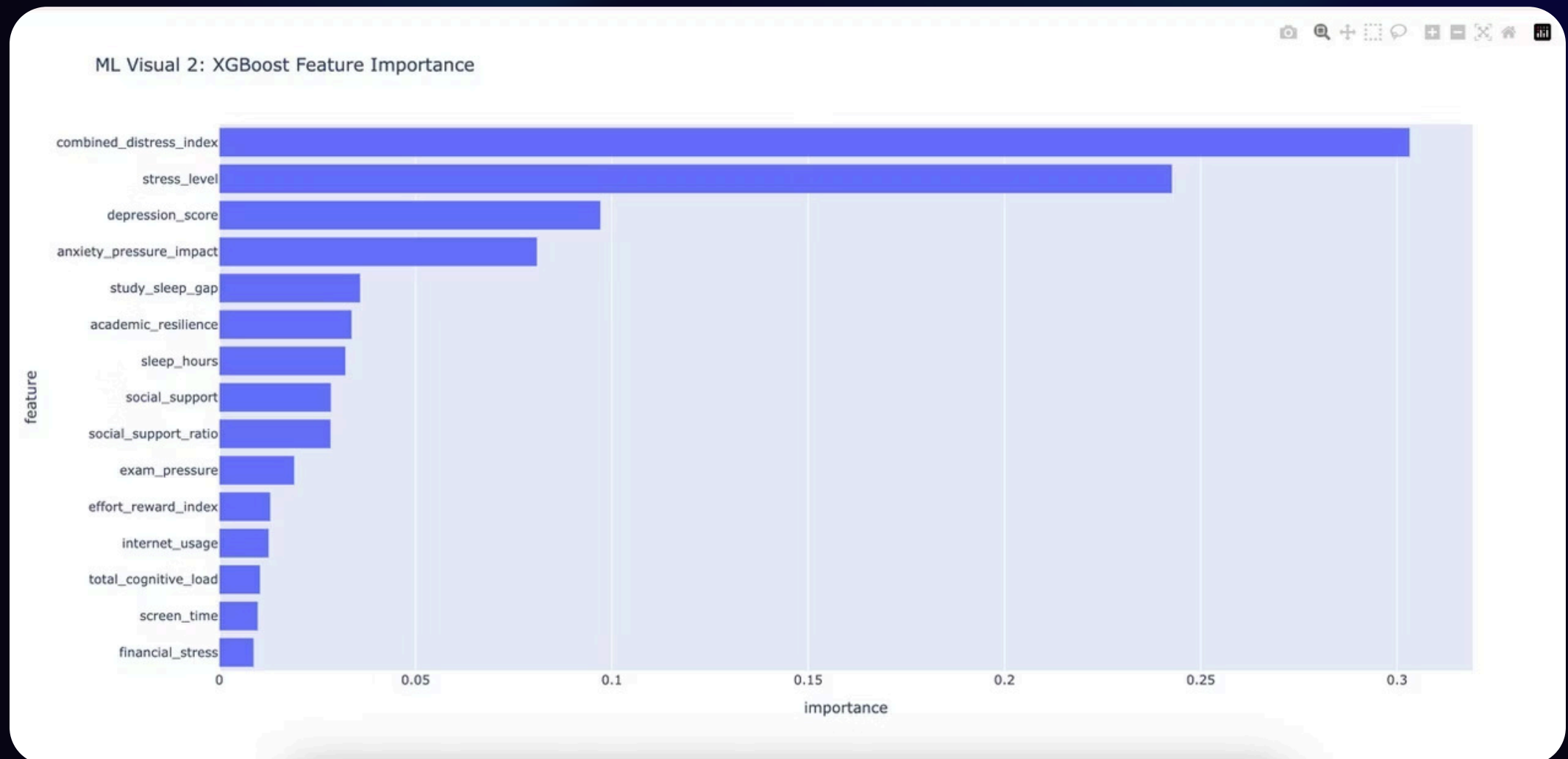
- Evaluates performance under class imbalance
- Focus on identifying high-risk students
- Maintains strong precision and recall



Feature Importance Analysis

- Ranks features by impact on predictions
- Top drivers: stress level, distress, resilience
- Engineered features improve performance

Key Insight → Burnout is driven by behavioral + psychological factors

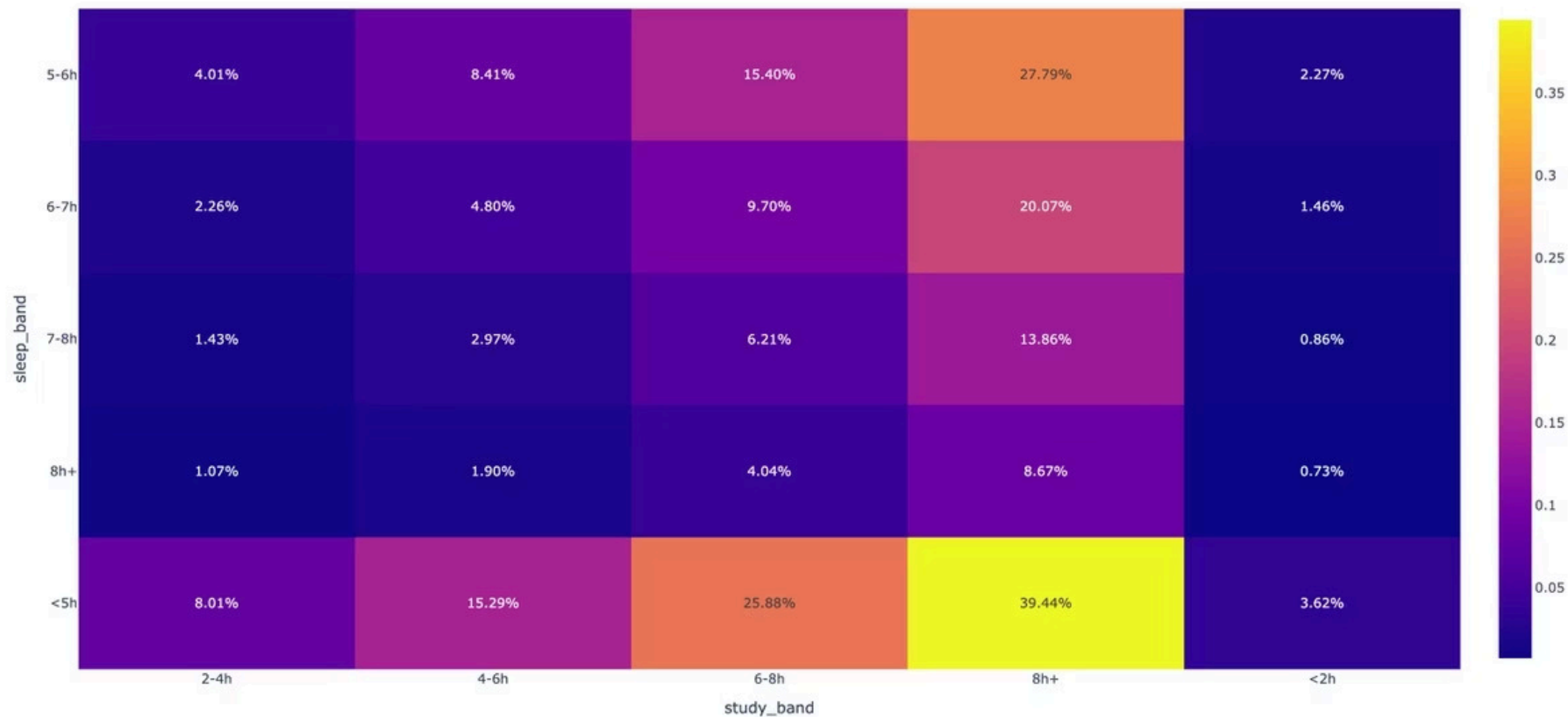


Behavioral Insight: Sleep vs Study Impact

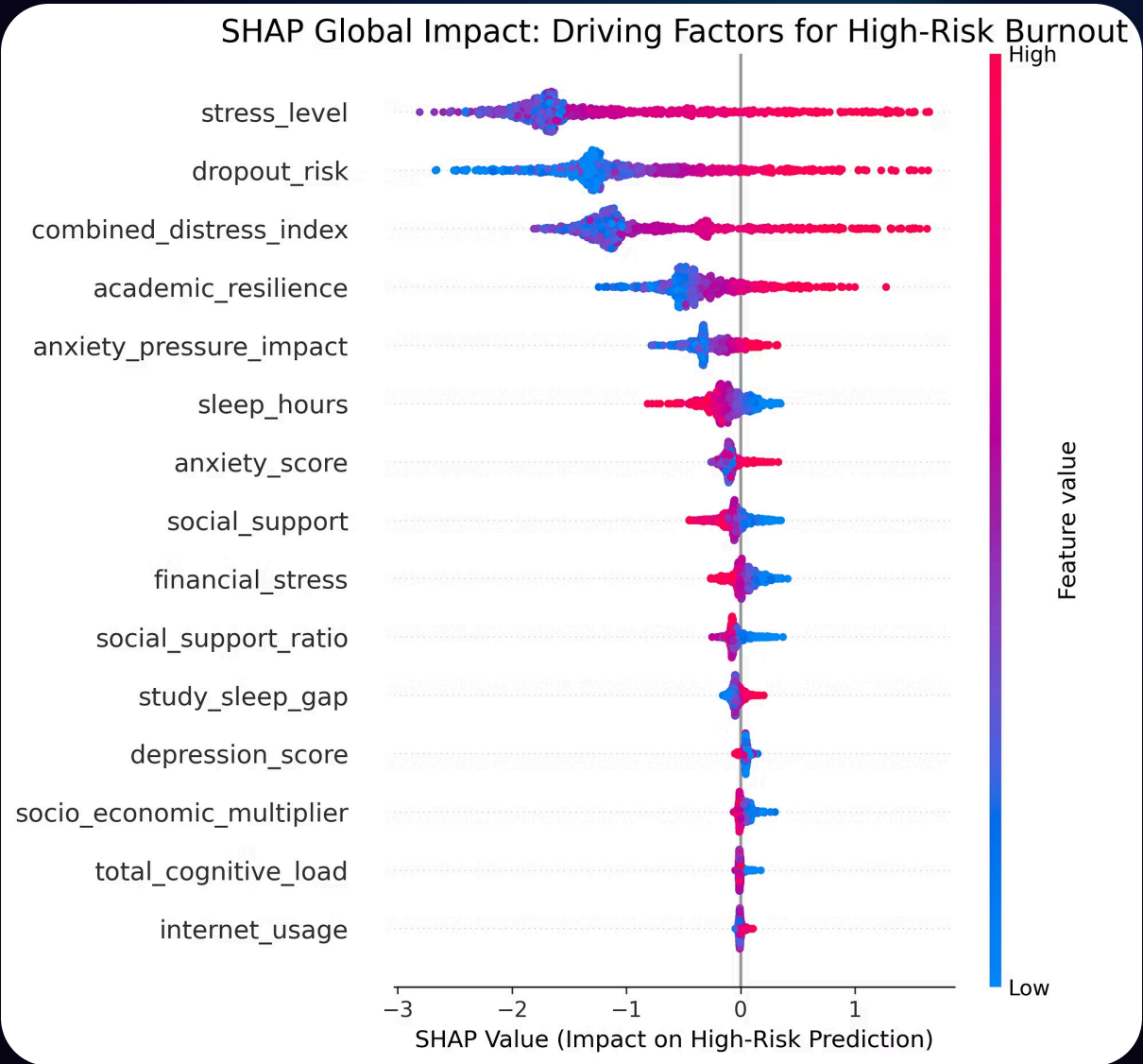
- Highest burnout risk: low sleep + high study
- Confirms interaction effects
- Provides actionable insight

Key Insight → Burnout is caused by imbalance between workload and recovery

ML Visual 6: Predicted High-Risk Probability by Sleep and Study Bands



Model Interpretability: Key Drivers of Burnout Risk



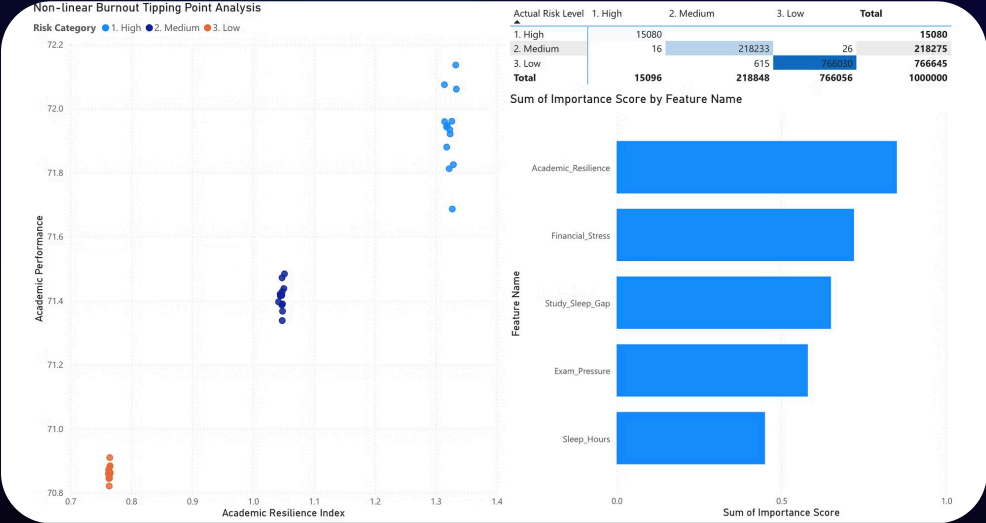
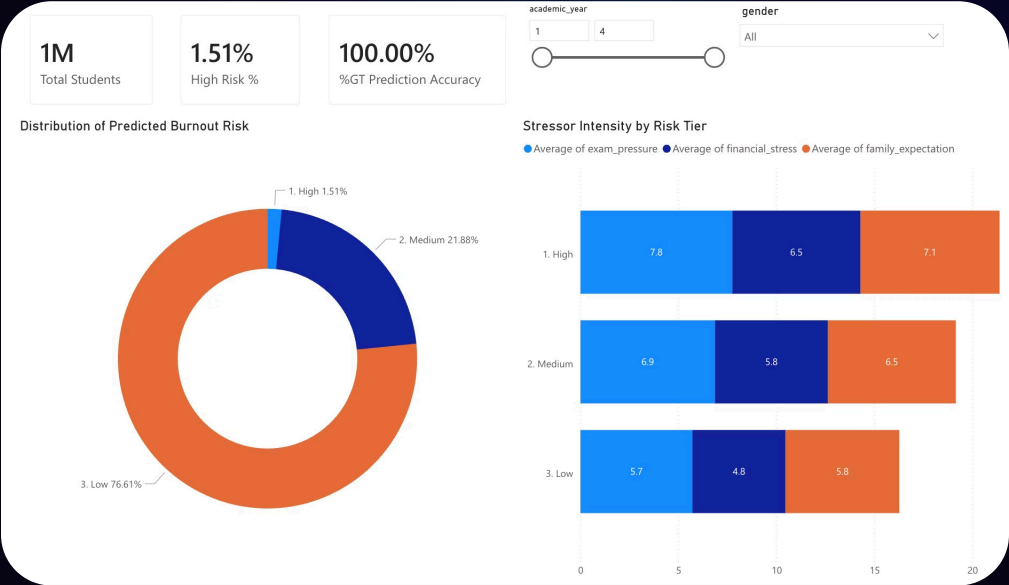
To understand how the model makes predictions, we used SHAP (Explainable AI).

- Stress level and combined distress are the strongest predictors of burnout risk
- Academic resilience and study-sleep gap capture interaction effects
- Higher stress and distress push predictions toward high-risk burnout
- Sleep hours and social support reduce burnout risk

Key Insight: → The model is interpretable and aligns with real-world behavioral patterns

Strategic Insight: The SHAP analysis identifies 'Academic Resilience' and 'Financial Multiplier' as the primary triggers. This enables administrators to move beyond GPA monitoring and target behavioral risk factors for earlier intervention

Power BI: Population-Level Burnout Insights



Streamlit: Real-Time Burnout Risk Prediction

Interactive Tool:

- Allows real-time burnout risk prediction for individual students
- Users input key variables such as sleep hours, stress level, and exam pressure





KEY INSIGHT

Conclusion & Key Takeaways

- Built a scalable, GPU-accelerated burnout prediction system
- Identified key drivers: stress, sleep imbalance, distress
- Model is both accurate and interpretable

Final Insight:

- Burnout is driven by imbalance between workload and recovery
- AI can support early detection and intervention

Contribution Evidence

GitHub repo: <https://github.com/xinghanchen-xc/DATA230-Final-Group-Project>

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103 \subsection{Behavioral Interaction Analysis}

104

105 Exploratory analysis using Plotly-based visualizations revealed key behavioralFigure \reffig{tableau-dashboard} presents a Tableau dashboard developed during the midterm phase. This dashboard captures relationships between study habitours, sleep patternduration, physical activity, depression levels, and burnout outcomres. The results indicate that burnout risk increases

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107 \begin{figure}[H]

108 \centering

109 \includegraphics[width= higher study hours, particularly when accompanied by reduced sleep duration.

110

111 Additionally, combined behavioral factors such as sleep deprivation and su\linewidth

112 \{figures/tableau-dashboard.png}

113 \caption{Tableau dashboard showing relationships between study behavior, sleep patterns, and burnout indicators.}

114 \end{figure}

115

116 The scatter plot shows a positive relationship between study hours and burnout score, suggested academic pressure were found top that excessive academic workload contributes significantly to elevated burnout levels. These patterns highlight the importance of considering inte burnout. The heatmap further reveals that combinations of low sleep duration effects rather than isolated variable and high study hours are associated with the highest burnout levels.

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118 \subsection{Key Insights from EDA}

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120 Several important patterns were identified:

121

122 \begin{itemize}

\item \textbf{Non-linear burnout threshold:} Burnout levels remain relatively stable until stress levels

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5th May, 7:44 pm
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figures/risk_heatmap.jpeg
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Contributors

Contributions per week to main, excluding merge commits

Period: All Contributions: Commits

Commits over time

Weekly from Apr 18, 2026 to May 2, 2026

Contributor	Commits	++	--
xinghanchen-xc	42	62,141	1,374
Abhijitreddy	8	2,598	0
SanjReddyK	3	422	0

You: Sanjana Reddy Katham

Made with GAMMA

Thank you